

PREDICTIVE MODELLING OF STROKE RISK USING PROBABILISTIC AND NON-PROBABILISTIC CLASSIFICATION ALGORITHMS

TechRise Programme (AI/ML Track)

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1. INTRODUCTION

1.1 Background and Significance

Stroke continues to be one of the major public health problems worldwide because it causes death, long-term disability, loss of independence, and high treatment costs. Recent Global Burden of Disease estimates reported millions of new strokes and stroke-related deaths, with a large share of the burden occurring in low- and middle-income countries (Feigin et al., 2026). Many important stroke risk factors, including hypertension, high blood glucose, obesity, smoking, and heart disease, can be identified and managed before a stroke occurs.

The possibility of prevention makes early risk identification important. Machine learning models can examine several demographic, clinical, and lifestyle variables together and estimate which patients are more likely to experience stroke. Such models are not replacements for clinical assessment or brain imaging. Instead, they can serve as screening or decision-support tools, especially where specialist services and advanced investigations are limited.

Classification methods used for stroke prediction can be grouped broadly into probabilistic and non-probabilistic approaches. Logistic Regression and Naive Bayes directly estimate class probabilities using statistical assumptions. Support Vector Machine learns a separating boundary between classes, while K-Nearest Neighbours classifies a patient by examining similar patients in the training data. Because the methods learn in different ways, their results may differ even when they are trained on the same dataset.

Previous studies have not produced one consistent answer about which family of models performs best. Differences in preprocessing, imbalance handling, feature selection, threshold choice, and evaluation measures can change the apparent winner. Therefore, this study compares four algorithms using the same dataset, the same train-test split, and a common set of evaluation measures.

1.2 Statement of the Problem

Stroke datasets are usually highly imbalanced: most records belong to patients without stroke, while only a small minority belong to patients with stroke. In such a dataset, a model may obtain very high accuracy simply by predicting that almost everybody will not have a stroke. This can be dangerous because the model may miss the patients who actually need attention.

There is also a need to understand why a model was selected, not only to report its final score. A useful comparison should explain the preprocessing steps, hyperparameters, cross-validation choices, probability thresholds, confusion matrices, and practical costs of false-negative and false-positive predictions. This report therefore evaluates both predictive performance and practical suitability for stroke-risk screening.

1.3 Aim and Objectives

Aim: To compare probabilistic and non-probabilistic classification algorithms for predicting stroke using the Health Stroke Dataset, and to explain the reason for each major analytical decision.

The specific objectives are to:

1. Prepare the dataset by checking missing values, duplicate records, outliers, skewness, categorical variables, and class imbalance.
2. Develop and tune Logistic Regression and Mixed Naive Bayes as the probabilistic models.
3. Develop and tune Support Vector Machine and K-Nearest Neighbours as the non-probabilistic models.
4. Evaluate the four models using accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC.
5. Examine how changing the probability threshold affects the ability of each model to identify stroke cases.
6. Interpret the Logistic Regression coefficients, odds ratios, confidence intervals, and p-values.
7. Identify the best model by ROC-AUC and recommend the most suitable model for practical stroke-risk screening.

2. DATA PREPROCESSING

2.1 Dataset Description

The submitted program loaded the Health Stroke Dataset from a comma-separated values file. The execution produced an 80:20 split containing 4,088 training observations and 1,022 test observations, giving a total of 5,110 records. Each record represented one person. The outcome variable, stroke, was coded as 1 when stroke occurred and 0 when stroke did not occur.

Variable	Type	Meaning and modelling decision
id	Identifier	A record number. It was removed before modelling because it does not describe a clinical risk factor.
gender	Categorical	Patient's recorded gender.
age	Continuous	Age of the patient.
hypertension	Binary	Whether the patient had hypertension.
heart_disease	Binary	Whether the patient had a history of heart disease.
ever_married	Categorical/Binary	Whether the patient had ever been married.
work_type	Categorical	Type of employment or work status.
Residence type	Categorical/Binary	Urban or rural residence.
avg_glucose_level	Continuous	Recorded average glucose level.
bmi	Continuous	Body mass index.
smoking_status	Categorical	Smoking history or current smoking category.
stroke	Binary outcome	Target variable: 1 for stroke and 0 for no stroke.

Table 1. Variables used in the stroke prediction analysis.

2.2 Data Preprocessing

2.2.1 Creation of Separate Working Copies

The original dataframe was copied before preprocessing. A second copy was also created for Mixed Naive Bayes. This decision was necessary because the models did not require exactly the same data representation. Logistic Regression, SVM, and KNN used one-hot encoded features and standardized values. Mixed Naive Bayes required the categorical variables to remain identifiable as categorical columns.

2.2.2 Missing-Value Treatment

The code checked every column for missing values. Missing BMI values were replaced with the median BMI. The median was chosen because BMI may contain unusually high or low values, and the median is less affected by such values than the mean. This allowed all observations to remain in the analysis instead of deleting patients with missing BMI.

2.2.3 Duplicate Records

The program checked for duplicated rows. Duplicate checking was included because repeated patient records can make a model appear more accurate than it really is. If the same or nearly identical record appears in both training and testing data, the test set is no longer fully independent. There were no duplicates.

2.2.4 Outlier Assessment

Age, average glucose level, and BMI were examined for outliers using the interquartile range method. For each variable, the first quartile and third quartile were calculated, and observations below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ were flagged. Boxplots were also produced.

The code did not automatically delete the flagged values. This was a reasonable decision because a very high glucose level or BMI may represent a genuine high-risk patient rather than a data-entry error. Removing such patients could remove clinically important information. Outliers should only be deleted after confirming that they are impossible or incorrectly recorded.

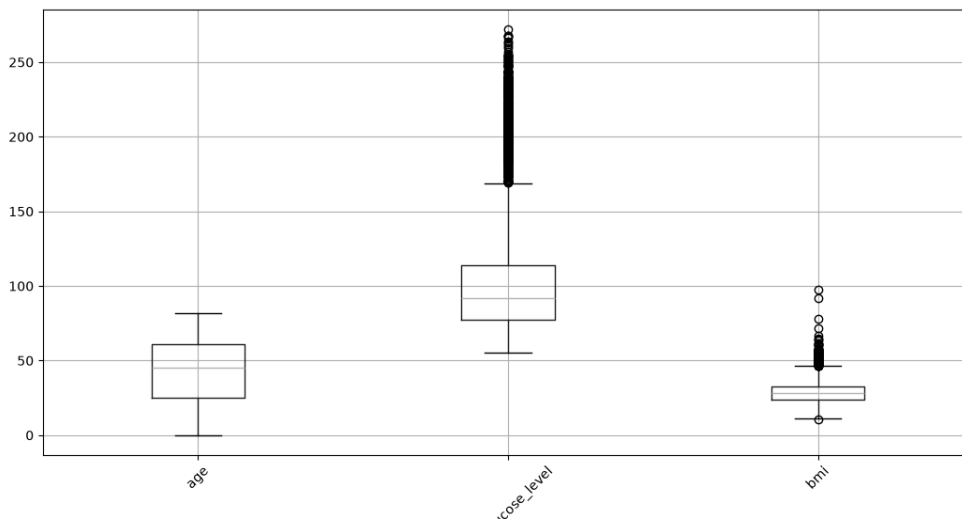


Figure 1. Boxplots of age, average glucose level, and body mass index before transformation.

Figure 1 shows that age had no prominent extreme observations outside the whiskers, whereas average glucose level had many high-value observations and BMI also had several high-value observations. The concentration of glucose observations above the upper whisker confirms that average glucose level was strongly right-skewed. These values were not automatically removed because they may represent genuine high-risk patients rather than errors. The graph therefore supported transforming average glucose level and using a robust median value when imputing missing BMI.

2.2.5 Skewness and Log Transformation

The skewness of age, average glucose level, and BMI was assessed. Average glucose level was then transformed using $\log(1 + \text{glucose})$. The $\log1p$ function was used because it reduces strong

right skew while remaining defined when a value is zero. Reducing skew can make linear and distance-based models less dominated by a small number of very large glucose values.

The transformation changes interpretation. In the inferential Logistic Regression model, the glucose coefficient represents a one-unit increase in $\log(1 + \text{glucose})$, not a one-unit increase in the original glucose measurement.

2.2.6 Encoding of Categorical Variables

For Logistic Regression, SVM, and KNN, gender, ever-married status, and residence type were converted to numbers using label encoding. Work type and smoking status were converted using one-hot encoding, with the first category dropped. Dropping one category prevents perfect duplication among dummy variables and gives the remaining categories a reference group for Logistic Regression.

For Mixed Naive Bayes, all categorical predictors were label encoded and their column positions were supplied to the model through the `categorical_features` argument. This allowed the model to treat those variables as categorical rather than as normally distributed continuous measurements. Gender was handled as a binary column in the submitted code. If the dataset contains more than two gender categories, one-hot encoding would be safer because label encoding can create an artificial numerical order. This should be checked in the final implementation.

2.2.7 Correlation Analysis

The identifier column was removed before calculating the correlation matrix. This was done because an ID number has no clinical meaning and any correlation with stroke would be accidental. Correlations were used for preliminary exploration only. A low correlation does not automatically mean that a variable is useless because a model may still use non-linear patterns or combinations of variables.

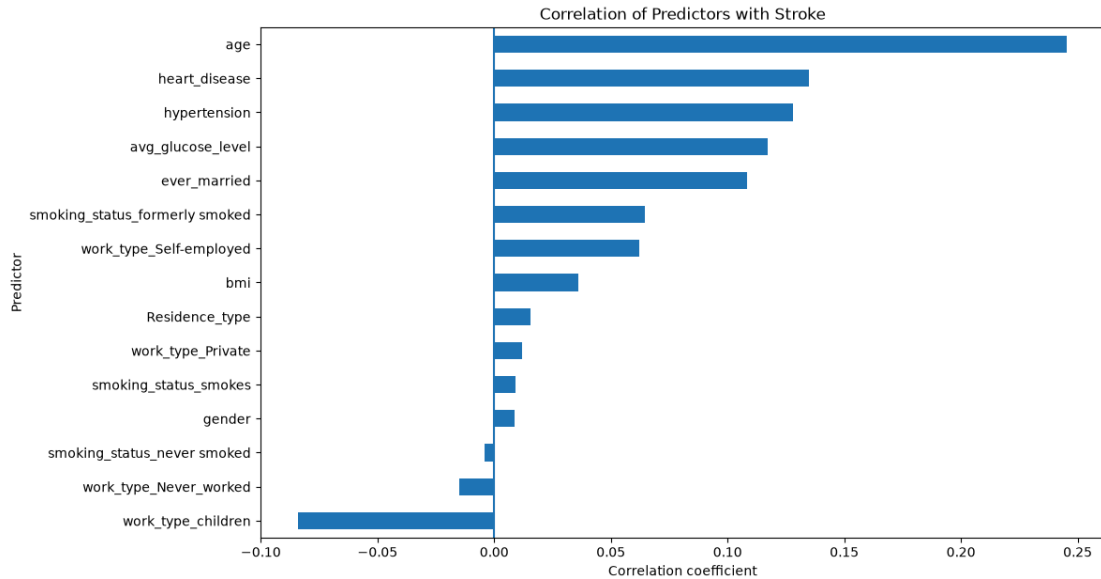


Figure 2. Pearson correlation coefficients between the encoded predictors and stroke.

Figure 2 shows that age had the strongest positive individual correlation with stroke, at approximately 0.25. Heart disease, hypertension, average glucose level, and ever-married status also had positive correlations, although all were weaker than age. The work-type category for children had the strongest negative correlation. These coefficients describe simple one-variable linear relationships only. They do not prove causation and should not be used alone to decide which variables enter the prediction models.

2.2.8 Train-Test Split

The data were divided into 80% training data and 20% testing data. The random_state was fixed at 42, making the split reproducible. Stratification was applied using the stroke outcome. Stratification was important because stroke cases were rare, and a random split without stratification could place too many or too few stroke cases in one partition.

The test set contained 972 non-stroke cases and 50 stroke cases. Therefore, only 4.89% of the test observations were stroke cases. This confirms that the data were severely imbalanced.

2.2.9 Feature Scaling

A StandardScaler was fitted on the training predictors and then applied to both the training and test predictors. Fitting the scaler only on the training set was a correct decision because it prevented the test-set mean and standard deviation from influencing model training.

Scaling was especially important for SVM and KNN. SVM uses distances and margins, while KNN directly compares distances between observations. Without scaling, a variable with a large numerical range could dominate the distance calculation. Scaling also helped the numerical optimisation of Logistic Regression.

2.2.10 Handling Class Imbalance

The program handled imbalance in three main ways: stratified splitting, balanced class weights for the selected Logistic Regression and SVM models, and probability-threshold analysis.

Balanced class weights gave more importance to errors involving the rare stroke class. ROC-AUC was used during tuning because it evaluates ranking ability across thresholds and is less misleading than accuracy in an imbalanced dataset. Threshold analysis examined whether lowering the decision threshold could identify more stroke cases.

Decision	Why it was taken
Median BMI imputation	Retained incomplete records and reduced the influence of extreme BMI values.
IQR outlier check	Flagged unusual values without automatically deleting clinically meaningful high-risk cases.
log1p glucose transformation	Reduced right skew and limited the influence of very high glucose values.
Label and one-hot encoding	Converted text categories into numerical form suitable for the selected algorithms.
80:20 stratified split	Preserved the rare stroke proportion while keeping an independent test set.
Training-only StandardScaler	Prevented test-set scaling information from entering model training.
Balanced class weights	Increased the cost of misclassifying the minority stroke class.
ROC-AUC tuning	Avoided selecting models mainly because of high majority-class accuracy.
Threshold analysis	Showed the trade-off between finding more stroke cases and producing more false alarms.

Table 2. Summary of preprocessing and evaluation decisions.

3. Model Development and Hyperparameter Tuning

3.1 Logistic Regression

Logistic Regression estimates the probability of stroke by modelling the log-odds of the outcome as a linear combination of the predictors. It was selected because it is widely used, produces probability estimates, and allows direct interpretation through coefficients and odds ratios.

GridSearchCV tested several solvers, regularisation methods, C values, and class-weight settings. Five-fold cross-validation was used. Five folds give a reasonable balance between

stability and computing time. ROC-AUC was the scoring measure because the outcome was highly imbalanced.

The best combination was $C = 0.01$, `class_weight = balanced`, elastic-net regularisation, `l1_ratio = 0.9`, and the saga solver. A small C means strong regularisation. Strong regularisation was selected because it produced the highest mean cross-validated ROC-AUC and can reduce overfitting. An `l1_ratio` of 0.9 makes the penalty mostly L1 with a smaller L2 component, encouraging a simpler model while still stabilising correlated predictors. Saga was necessary because it supports elastic-net regularisation.

The best cross-validation ROC-AUC was 0.84199. The output contained warnings because `l1_ratio` was also passed to some non-elastic-net candidates, where it is not used. Some high- C saga candidates also reached the maximum number of iterations. These warnings do not automatically invalidate the winning combination, but the grid should be cleaned in future code so that only meaningful parameter combinations are tested.

3.2 Mixed Naive Bayes

Mixed Naive Bayes was used instead of a purely Gaussian Naive Bayes model because the dataset contained both continuous and categorical predictors. The model applied appropriate probability distributions to the different feature types.

The smoothing parameter α was tested at 0.01, 0.1, 0.5, 1, 5, and 10. Smoothing prevents categories with very small or zero counts from producing zero probabilities. $\alpha = 5$ gave the highest reported ROC-AUC of 0.81335 and was therefore selected.

3.3 Support Vector Machine

SVM was selected because it can find a decision boundary with a wide margin and can model both linear and non-linear relationships through kernels. The grid tested linear and radial-basis-function kernels, C values of 0.1, 1, and 10, selected gamma values for the RBF kernel, and both ordinary and balanced class weights.

Three-fold cross-validation was used for SVM rather than five folds. This reduced computing time because SVM fitting and probability estimation were more expensive. The best setting was a linear kernel, $C = 0.1$, and balanced class weights, with a cross-validation ROC-AUC of 0.83519. The linear kernel was selected because it outperformed the tested RBF combinations; the more complex boundary was not needed for this dataset.

The model was refitted with probability output enabled so that ROC-AUC and threshold analysis could be calculated.

3.4 K-Nearest Neighbours

KNN was included as a distance-based, non-parametric method. The initial model used five neighbours, distance weighting, and Minkowski distance. Hyperparameter tuning tested 3, 5, 7, 9, 11, 15, and 21 neighbours; uniform and distance weights; and Euclidean, Manhattan, and Minkowski distance.

Five-fold cross-validation with ROC-AUC scoring selected 21 neighbours, uniform weights, and Euclidean distance. The larger number of neighbours produced a smoother and less variable decision rule. Nevertheless, at the ordinary decision rule, the tuned model predicted no stroke cases. This demonstrates that a statistically smoother neighbourhood was still dominated by the majority class.

Model	Selection method	Selected settings	Selection score
Logistic Regression	5-fold GridSearchCV; ROC-AUC	C=0.01; balanced; elastic net; l1_ratio=0.9; saga	0.84199
Mixed Naive Bayes	Manual alpha comparison	alpha=5	0.81335 on test set
SVM	3-fold GridSearchCV; ROC-AUC	linear kernel; C=0.1; balanced	0.83519
KNN	5-fold GridSearchCV; ROC-AUC	k=21; uniform; Euclidean	0.72545

Table 3. Hyperparameter tuning decisions and selected settings.

3.5 Evaluation Measures

Measure	Meaning	Reason for use
Accuracy	The proportion of all test observations classified correctly.	Useful as a general summary, but easily inflated by the large non-stroke class.
Precision	The proportion of predicted stroke cases that were truly stroke cases.	Shows how many stroke alerts were correct.
Recall/Sensitivity	The proportion of actual stroke cases correctly identified.	Very important in screening because a false negative means a stroke case was missed.
F1-score	The harmonic mean of precision and recall.	Balances the need to identify stroke cases with the need to control false alarms.
ROC-AUC	The probability that the model ranks a randomly selected stroke case above a randomly selected non-stroke case.	Provides a threshold-independent measure of discrimination.

Table 4. Model evaluation measures.

4. RESULTS AND DISCUSSION

4.1 Test-Set Class Distribution and the Accuracy Problem

The test set contained 1,022 observations: 972 without stroke and 50 with stroke. A model that predicted 'no stroke' for every patient would obtain an accuracy of $972/1,022 = 95.11\%$ while having 0% recall for stroke. This no-information accuracy is an important reference point.

The tuned KNN model at its ordinary decision rule obtained exactly 95.11% accuracy and 0% recall because it predicted every patient as non-stroke. Therefore, the apparently high accuracy represented failure rather than strong clinical performance. This is why recall, F1-score, confusion matrices, and ROC-AUC were given greater importance than accuracy alone.

4.2 Overall Model Performance

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.7348	0.1353	0.82	0.2323	0.8406
Mixed Naive Bayes	0.9002	0.1977	0.34	0.2500	0.8134
Support Vector Machine	0.7319	0.1340	0.82	0.2303	0.8422
K-Nearest Neighbours	0.8581	0.1429	0.38	0.2077	0.7330

Table 5. Final comparison table produced by the submitted code. KNN uses its threshold-adjusted prediction at 0.10.

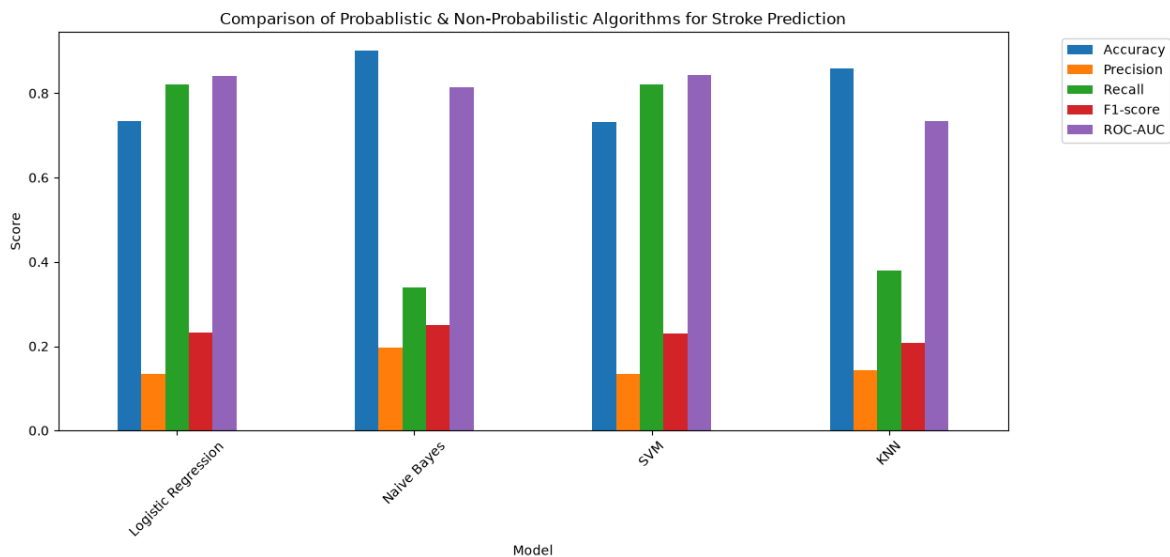


Figure 3. Final comparison of the four classification algorithms on the held-out test set.

Figure 3 makes the class-imbalance trade-off clear. Mixed Naive Bayes and KNN had higher accuracy because they classified most non-stroke observations correctly, but their recall for the stroke class was much lower. Logistic Regression and SVM had lower overall accuracy but each detected 82% of the actual stroke cases. SVM had the highest ROC-AUC, while Logistic Regression was almost identical and was easier to interpret. Therefore, the graph should not be read by selecting the tallest accuracy bar alone.

4.3 Model-Specific Results

The following subsections present the confusion matrices and ROC curves for each model. Confusion matrices explain the actual classification errors at a chosen threshold, while ROC curves summarise discrimination across all possible thresholds.

4.3.1 Logistic Regression Results

The tuned Logistic Regression model achieved an accuracy of 73.48%, precision of 13.53%, recall of 82.00%, F1-score of 0.2323, and ROC-AUC of 0.8406. It correctly identified 41 of the 50 stroke cases and missed 9. It also incorrectly labelled 262 non-stroke patients as stroke cases.

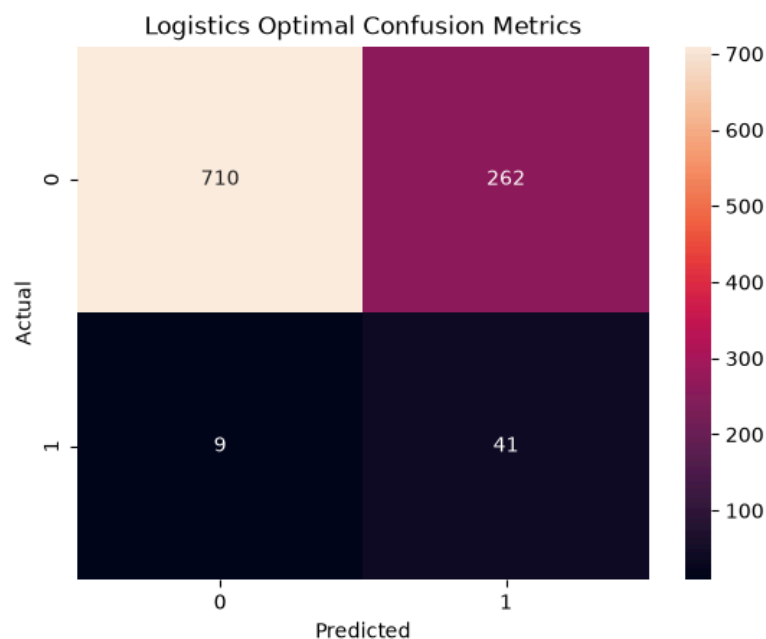


Figure 4. Confusion matrix of the tuned Logistic Regression model at the selected threshold of 0.50.

Figure 4 shows 710 true negatives, 262 false positives, 9 false negatives, and 41 true positives. Thus, the model identified 41 of the 50 stroke cases and missed 9. The 262 false-positive alerts explain the low precision. In a screening setting, this pattern may still be acceptable when the priority is to avoid missing patients who may require further clinical assessment.

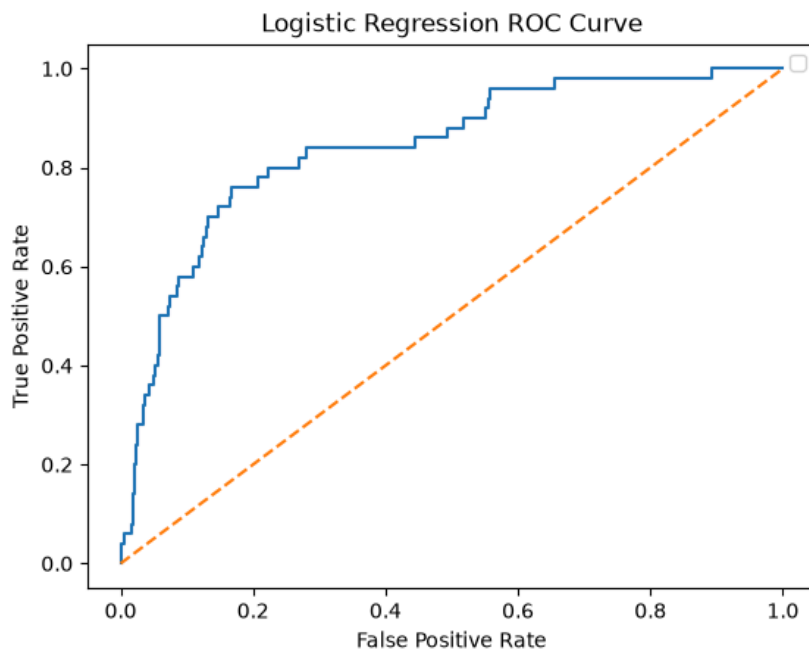


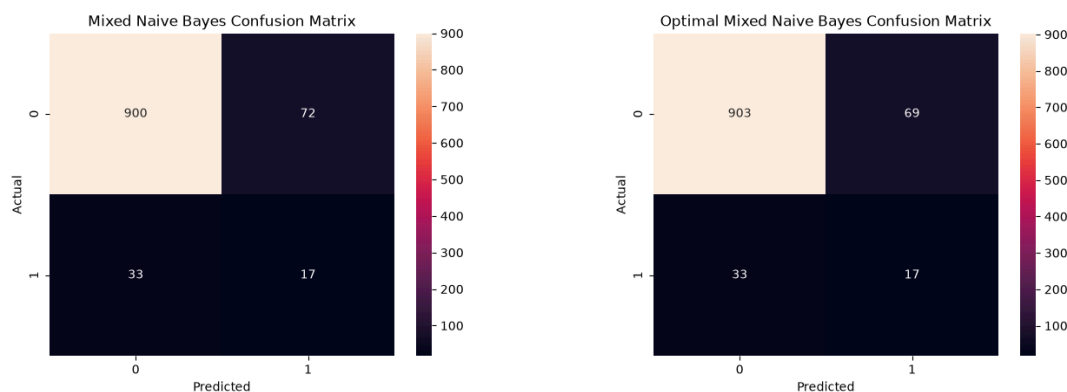
Figure 5. Receiver operating characteristic curve for Logistic Regression.

The ROC curve in Figure 5 remains well above the diagonal reference line for most thresholds. Its area under the curve was 0.8406, meaning that the model had good ability to rank a randomly selected stroke case above a randomly selected non-stroke case. The curve also shows that gaining very high sensitivity requires accepting a higher false-positive rate.

The relatively low precision means that most positive alerts were false positives. However, the high recall is valuable for screening because only 18% of the stroke cases were missed. In a first-stage screening system, false positives can be reviewed by a clinician, while a false negative may leave a high-risk patient without further assessment.

4.3.2 Mixed Naive Bayes Results

After alpha tuning, Mixed Naive Bayes achieved 90.02% accuracy, 19.77% precision, 34.00% recall, an F1-score of 0.2500, and ROC-AUC of 0.8134. It correctly identified 17 stroke cases and missed 33. It produced 69 false-positive alerts.



(a) Initial Mixed Naive Bayes model

(b) Tuned model with $\alpha = 5$

Figure 6. Confusion matrices before and after tuning Mixed Naive Bayes.

Figure 6 shows that tuning produced only a small improvement. The initial model correctly classified 900 non-stroke cases and produced 72 false positives, while the tuned model correctly classified 903 non-stroke cases and reduced false positives to 69. Both models detected 17 stroke cases and missed 33. Therefore, alpha tuning improved accuracy and precision slightly but did not improve sensitivity to the stroke class.

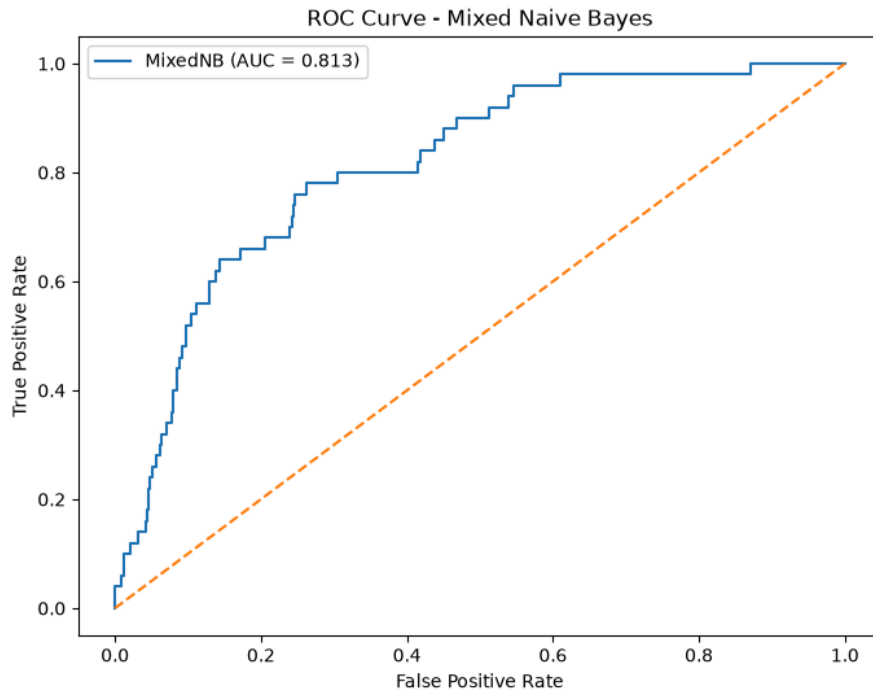


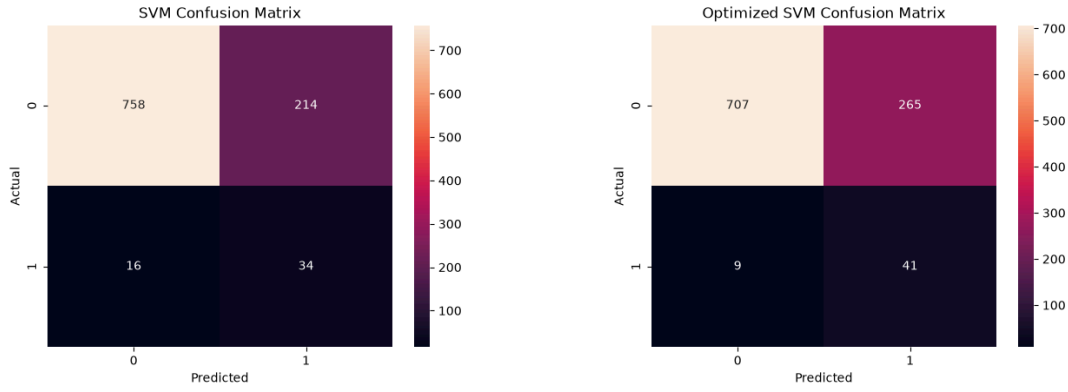
Figure 7. Receiver operating characteristic curve for the tuned Mixed Naive Bayes model.

Figure 7 shows an ROC-AUC of approximately 0.813. This indicates good, but weaker, discrimination than Logistic Regression and SVM. The curve rises above the chance line, showing that the model contains useful ranking information even though its classification at the ordinary threshold had low recall.

Naive Bayes had the highest precision and F1-score in the final comparison table and a much higher accuracy than Logistic Regression and SVM. However, its recall was low. It missed two-thirds of the stroke cases. For a stroke-screening task, this is a serious weakness even though the overall accuracy appears attractive.

4.3.3 Support Vector Machine Results

The tuned SVM achieved 73.19% accuracy, 13.40% precision, 82.00% recall, an F1-score of 0.2303, and ROC-AUC of 0.8422. It correctly detected 41 of the 50 stroke cases and missed 9, the same number as Logistic Regression. It generated 265 false positives.



(a) Initial RBF SVM

(b) Tuned linear SVM

Figure 8. Confusion matrices before and after SVM hyperparameter tuning.

Figure 8 demonstrates that tuning increased the number of detected stroke cases from 34 to 41 and reduced false negatives from 16 to 9. This gain in recall came with more false positives: the number increased from 214 to 265. The tuned balanced linear SVM therefore shifted the decision boundary toward the minority stroke class, which was appropriate for a screening-oriented objective.

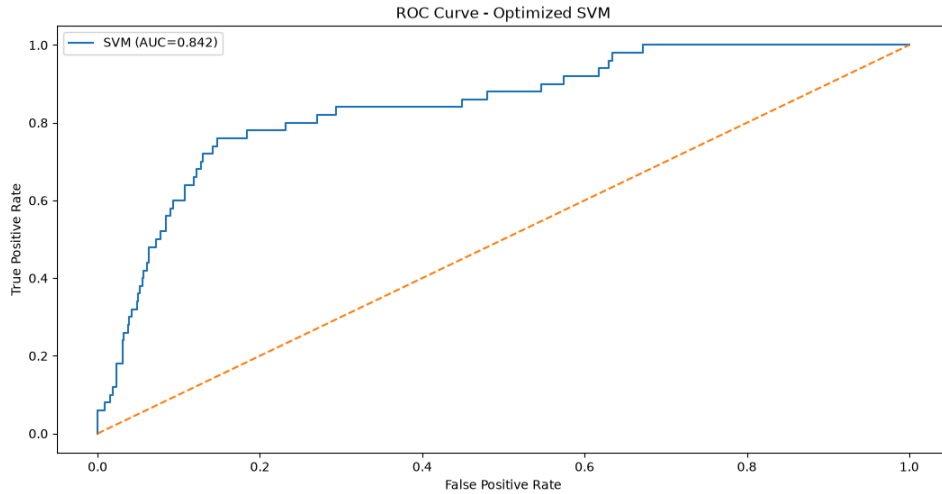


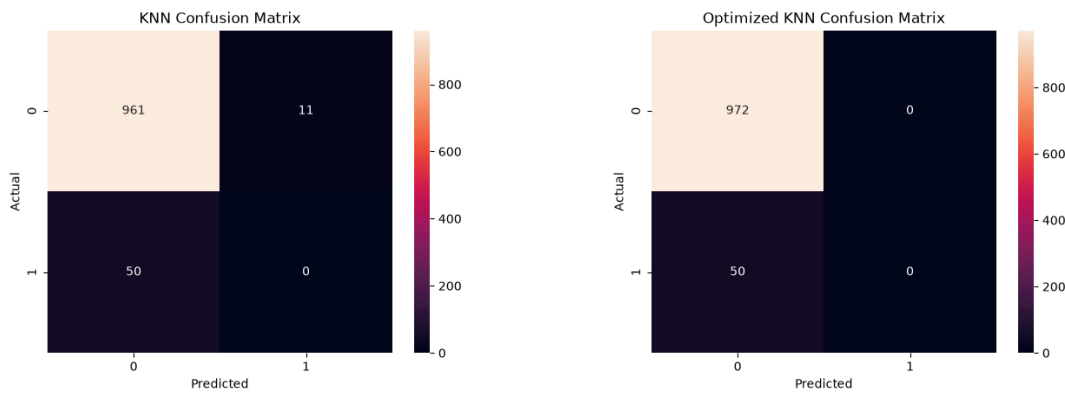
Figure 9. Receiver operating characteristic curve for the tuned SVM.

Figure 9 gives an ROC-AUC of 0.842, the highest value among the four models. The curve is consistently above the diagonal line and shows good separation of stroke and non-stroke observations. However, its numerical advantage over Logistic Regression was only about 0.002, so the graph does not establish a large practical difference between the two models.

SVM had the highest ROC-AUC of all four models, but its advantage over Logistic Regression was only 0.0016. No statistical test, such as a DeLong comparison of AUCs, was carried out. Therefore, the report should not claim that SVM was meaningfully superior; it was only the numerical winner by ROC-AUC in this test sample.

4.3.4 K-Nearest Neighbours Results

The initial KNN model achieved 94.03% accuracy but 0% recall and 0% F1-score. After hyperparameter tuning, the ordinary KNN classification rule achieved 95.11% accuracy, again with 0% recall. Both versions failed to identify any stroke case.



(a) Initial KNN model

(b) Tuned KNN at its ordinary decision rule

Figure 10. Confusion matrices of KNN before and after hyperparameter tuning.

Figure 10 shows the effect of severe class imbalance on KNN. The initial model produced 11 false-positive predictions but failed to identify any of the 50 stroke cases. After tuning, the model classified every observation as non-stroke, giving 972 true negatives and 50 false negatives. The resulting 95.11% accuracy was therefore misleading because recall for stroke was 0%.

A lower probability threshold of 0.10 was then used. With this adjusted threshold, KNN achieved 85.81% accuracy, 14.29% precision, 38.00% recall, an F1-score of 0.2077, and ROC-AUC of 0.7330. It correctly detected 19 stroke cases but still missed 31. The lower threshold improved minority-class detection but KNN remained the weakest model by ROC-AUC.

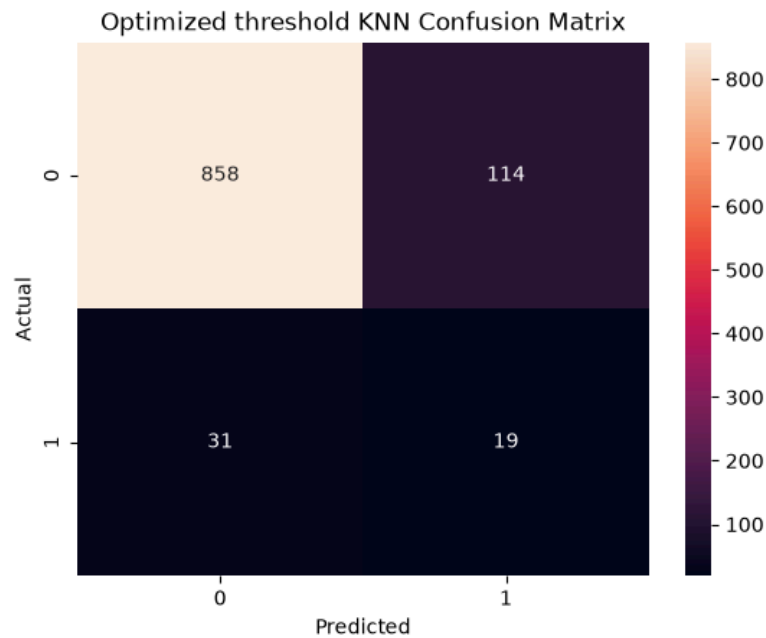


Figure 11. Confusion matrix of the tuned KNN model after reducing the classification threshold to 0.10.

Figure 11 shows that lowering the threshold allowed KNN to detect 19 of the 50 stroke cases. The model then produced 858 true negatives, 114 false positives, 31 false negatives, and 19 true positives. Although recall improved from 0% to 38%, the model still missed most stroke cases and remained weaker than Logistic Regression and SVM.

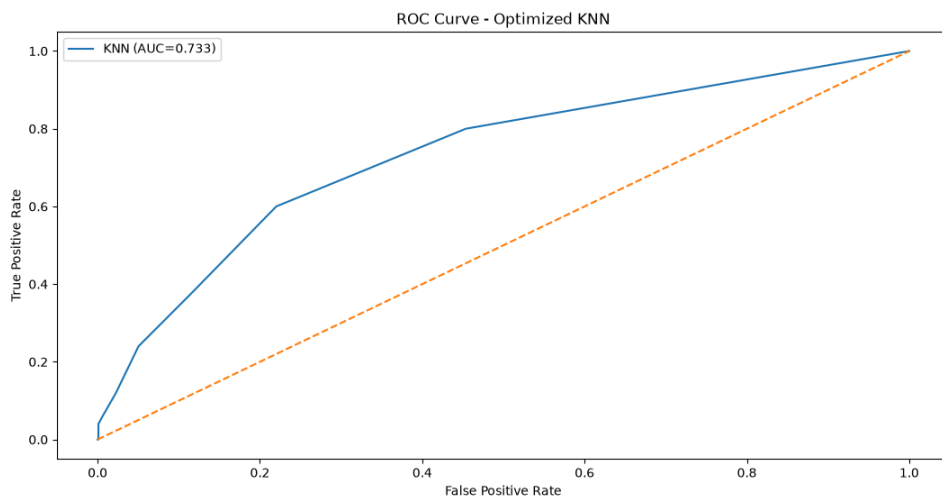


Figure 12. Receiver operating characteristic curve for the tuned KNN model.

Figure 12 shows an ROC-AUC of approximately 0.733. The model performed better than random guessing, but its curve was closer to the diagonal reference line than the curves of the

other models. This confirms that KNN had the weakest ability to separate stroke from non-stroke observations in the present analysis.

Model	True negatives	False positives	False negatives	True positives	Meaning
Logistic Regression	710	262	9	41	41/50 stroke cases detected
Mixed Naive Bayes	903	69	33	17	Best precision, but 33 strokes missed
SVM	707	265	9	41	Highest AUC; same stroke detection as Logistic Regression
KNN, threshold 0.10	858	114	31	19	Threshold helped, but recall remained low

Table 6. Confusion-matrix counts for the final predictions.

4.4 Threshold Analysis

A classification threshold determines the probability at which a patient is labelled as a stroke case. The default threshold is usually 0.50. Lowering the threshold generally increases recall because more patients are labelled positive, but it also increases false positives and reduces precision.

Threshold	Accuracy	Precision	Recall	F1-score
0.05	0.1683	0.0546	0.98	0.1034
0.10	0.3053	0.0646	0.98	0.1213
0.15	0.3885	0.0715	0.96	0.1331
0.20	0.4589	0.0801	0.96	0.1479
0.25	0.5117	0.0819	0.88	0.1499
0.30	0.5656	0.0896	0.86	0.1623
0.35	0.5998	0.0948	0.84	0.1704
0.40	0.6517	0.1077	0.84	0.1909
0.45	0.6947	0.1214	0.84	0.2121
0.50	0.7348	0.1353	0.82	0.2323

Table 7. Logistic Regression threshold analysis.

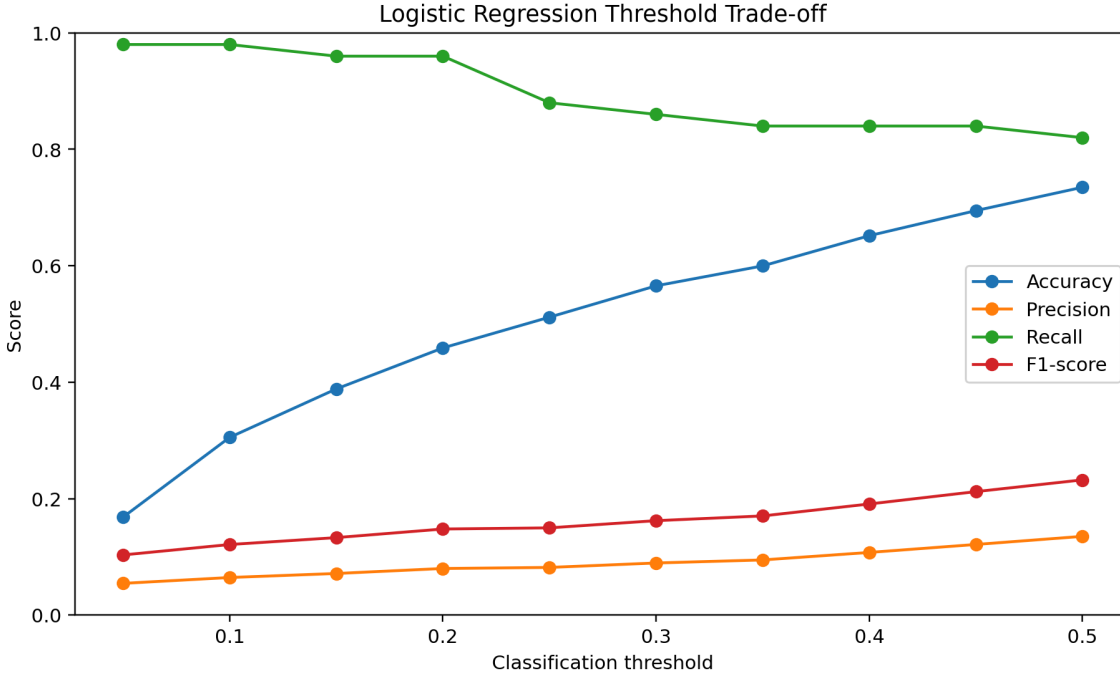


Figure 13. Effect of changing the Logistic Regression probability threshold.

Figure 13 shows the expected threshold trade-off. At very low thresholds, recall approached 98%, but accuracy and precision fell sharply because many non-stroke patients were labelled positive. As the threshold increased, accuracy, precision, and F1-score improved, while recall decreased gradually. Among the tested values, 0.50 produced the highest F1-score and was retained for the reported Logistic Regression confusion matrix.

For Logistic Regression, thresholds of 0.05 and 0.10 detected 98% of stroke cases, but precision was only 5.46% and 6.46%. This means a very large number of non-stroke patients would be sent for further assessment. Among the tested thresholds, 0.50 produced the highest F1-score. It was therefore retained in the submitted comparison.

For Mixed Naive Bayes, the highest reported F1-score occurred at a threshold of 0.30, where recall was 54% and F1 was 0.30. For SVM, a probability threshold of 0.10 produced recall of 72% and F1 of approximately 0.336. For KNN, a threshold of 0.10 produced its best reported F1-score of approximately 0.208.

SVM requires an additional caution. The class returned by `SVC.predict` is based on the SVM decision function, while the probability returned by `predict_proba` is obtained through probability calibration. Therefore, the model's direct class prediction is not guaranteed to be identical to

applying a 0.50 threshold to its probability. This explains why the SVM threshold output differed from the direct SVM prediction.

4.5 Logistic Regression Statistical Interpretation

A separate statsmodels Logistic Regression was fitted on the unscaled training data to examine coefficients, odds ratios, confidence intervals, and p-values. This explanatory model was different from the tuned predictive Logistic Regression because it did not use elastic-net regularisation or balanced class weights. The two models answered different questions: the scikit-learn model focused on prediction, while the statsmodels model focused on association and interpretation.

The overall likelihood-ratio test was statistically significant ($p < 0.001$), indicating that the predictors jointly improved the model compared with an intercept-only model. McFadden's pseudo- R^2 was 0.201. This is not the same as the ordinary R^2 used in linear regression; it should be treated as a measure of improvement in model likelihood, not as the percentage of stroke explained.

Predictor	Coefficient	Odds ratio	95% CI for OR	p-value	Conclusion
Gender code	0.0560	1.058	0.776-1.442	0.723	Not significant
Age	0.0718	1.074	1.061-1.088	<0.001	Significant positive association
Hypertension	0.4732	1.605	1.122-2.296	0.010	Significant positive association
Heart disease	0.2315	1.260	0.826-1.923	0.283	Not significant after adjustment
Ever married	-0.2877	0.750	0.462-1.218	0.245	Not significant
Urban residence code	0.0870	1.091	0.807-1.476	0.572	Not significant
Log-transformed glucose	0.4581	1.581	1.106-2.259	0.012	Significant positive association
BMI	-0.0012	0.999	0.974-1.023	0.921	Not significant
Never worked	-0.1200	0.887	0.000-31476.610	0.982	Very imprecise; sparse category
Private work	0.0415	1.042	0.671-1.619	0.853	Not significant
Self-employed	-0.2359	0.790	0.481-1.297	0.351	Not significant
Children work category	-1.6504	0.192	0.002-23.104	0.499	Very imprecise; sparse stroke cases
Formerly smoked	0.0987	1.104	0.706-1.725	0.665	Not significant
Never smoked	-0.1233	0.884	0.578-1.353	0.570	Not significant
Currently smokes	0.1884	1.207	0.731-1.996	0.463	Not significant

Table 8. Logistic Regression coefficient and odds-ratio interpretation. Reference categories are determined by the encoding used in the code.

Age was the strongest clearly significant predictor. The odds ratio was approximately 1.074 per additional year of age, meaning that the odds of stroke increased by about 7.4% for each additional year, holding the other variables constant. Over ten years, this corresponds to roughly twice the odds because 1.074^{10} is about 2.05.

Hypertension had an odds ratio of approximately 1.605. Patients coded as hypertensive had about 60.5% higher odds of stroke than patients without hypertension, after adjustment for the other predictors. The 95% confidence interval was approximately 1.122 to 2.296 and did not include 1.

Log-transformed average glucose level had an odds ratio of approximately 1.581 per one-unit increase in $\log(1 + \text{glucose})$. This should not be described as a 58% increase for every one-unit increase in the original glucose measurement. A more understandable interpretation is that an approximate doubling of $(1 + \text{glucose})$ corresponds to about 37% higher odds of stroke, holding other predictors constant.

The other predictors were not statistically significant at the 5% level after adjustment. This does not prove that they have no clinical relationship with stroke. A non-significant result can occur because the effect is small, categories are sparse, variables overlap with one another, or the dataset contains too few stroke cases. The extremely wide confidence intervals for the 'never worked' and 'children' categories show that those estimates were unstable.

4.6 Direct Comparison of the Four Models

Decision criterion	Selected model	Reason
Highest ROC-AUC	SVM (0.8422)	It ranked stroke and non-stroke cases slightly better than the other models.
Best probabilistic screening model	Logistic Regression	AUC 0.8406, recall 82%, transparent coefficients, and slightly better precision/F1 than SVM.
Highest accuracy in final table	Mixed Naive Bayes (90.02%)	It made many correct majority-class predictions, but detected only 34% of stroke cases.
Highest precision and F1 in final table	Mixed Naive Bayes	Precision 19.77% and F1 0.25, but this came with low recall.
Highest recall	Logistic Regression and SVM (82%)	Each detected 41 of the 50 stroke cases.
Weakest discrimination	KNN (AUC 0.7330)	It was strongly affected by class imbalance and initially predicted no strokes.
Most interpretable	Logistic Regression	Effects can be explained using coefficients, odds ratios, confidence intervals, and p-values.

Table 9. Model-selection decisions under different criteria.

By the formal criterion used in the code, SVM was the best model because it had the highest ROC-AUC. However, the difference between SVM and Logistic Regression was very small. Logistic Regression had the same recall, slightly higher precision and F1-score, and much better interpretability. Therefore, the two models should be regarded as practically close rather than as a clear winner and loser.

For a clinical screening application, Logistic Regression is the preferred model from the current results. It detected 82% of stroke cases and produced a probability that can be explained to clinicians. SVM is a reasonable alternative when ranking performance is the main objective and interpretability is less important.

Mixed Naive Bayes may be useful where fewer false alerts are preferred, but its low recall makes it unsuitable as the only screening model. KNN should not be selected for deployment based on the current analysis because its standard prediction rule failed to identify any stroke case and its threshold-adjusted discrimination remained the weakest.

4.7 Technical Warnings and Methodological Limitations

- The dataset was highly imbalanced, with only 50 stroke cases in the 1,022-record test set. Performance estimates for the stroke class therefore have considerable uncertainty.
- Median imputation and categorical encoding were performed before the train-test split. A production pipeline should fit all learned preprocessing steps on the training data only.
- Mixed Naive Bayes alpha and the model thresholds were selected using the test set. This makes the reported threshold-optimised results optimistic.
- Different cross-validation fold counts were used: five folds for Logistic Regression and KNN, but three folds for SVM. This was computationally reasonable but makes the stability of the tuning scores slightly different.
- The final comparison mixed default/direct predictions for three models with a threshold-adjusted prediction for KNN. A fully fair comparison should apply one threshold-selection rule to all models.
- The probability output of SVM was produced by SVC probability calibration, and the terminal output showed a deprecation warning. `CalibratedClassifierCV` should be used in updated code.
- Some Logistic Regression saga candidates did not converge, and the parameter grid contained redundant `l1_ratio` combinations. The grid should be corrected and convergence checked explicitly.
- The study used one train-test split. Repeated stratified cross-validation, nested cross-validation, or external validation would provide more reliable estimates.

- No confidence intervals were calculated for the predictive metrics, and no statistical test compared the AUCs. Small numerical differences should therefore be interpreted cautiously.
- The model is a research model and should not be used to diagnose stroke or replace clinical assessment.
- One exported image labelled as a final Mixed Naive Bayes confusion matrix contained the Logistic Regression counts (710, 262, 9, and 41), which conflicted with the reported Naive Bayes metrics. It was treated as a plotting or variable-reuse error and was not included as evidence. The correctly matched tuned Mixed Naive Bayes matrix was 903 true negatives, 69 false positives, 33 false negatives, and 17 true positives.

5 Conclusion

The four algorithms behaved differently because they responded differently to class imbalance and feature structure. SVM achieved the highest ROC-AUC of 0.8422, followed very closely by Logistic Regression at 0.8406. Both models identified 82% of the stroke cases. Mixed Naive Bayes had the highest final-table precision and F1-score but detected only 34% of stroke cases. KNN was heavily biased toward the majority class and had the lowest ROC-AUC.

The non-probabilistic family produced the numerical AUC winner through SVM, but this does not mean that the whole non-probabilistic family was better because KNN performed poorly. Similarly, the probabilistic models showed different trade-offs: Logistic Regression prioritised recall, while Mixed Naive Bayes produced fewer false positives but missed many stroke cases. Considering discrimination, recall, simplicity, and interpretability together, Logistic Regression is recommended as the most practical screening model from the present analysis. SVM remains the best model by the single numerical criterion of ROC-AUC.

5.1 Technical Recommendations

1. Use a Pipeline and ColumnTransformer so that imputation, encoding, and scaling are fitted only on the training data.
2. Select Naive Bayes alpha and all probability thresholds through cross-validation or a separate validation set, not the final test set.
3. Use one consistent comparison framework for all four models, including the same cross-validation strategy and threshold-selection rule.

4. Replace `SVC(probability=True)` with a properly calibrated SVM using `CalibratedClassifierCV`.
5. Report confidence intervals for recall, precision, F1-score, and ROC-AUC, and statistically compare the two leading AUCs.
6. Consider repeated stratified cross-validation and external validation on data from another hospital or population.
7. Assess probability calibration using a calibration curve, Brier score, and calibration intercept/slope before clinical use.
8. Choose the deployment threshold using clinical costs. Missing a high-risk patient may be more serious than generating an extra referral.
9. Retain Logistic Regression as the main interpretable model and compare it with SVM as a secondary high-discrimination model.
10. Do not deploy the model as a diagnostic tool. Use it only as decision support alongside clinical history, examination, and appropriate investigations.

5.2 Clinical Recommendations

1. Adopt Logistic Regression as the primary screening model because it provides the best balance between recall, ROC-AUC, simplicity and clinical interpretability.
2. Integrate the model as a clinical decision-support tool rather than a diagnostic replacement. Clinical assessment, examination and investigations should remain the responsibility of healthcare professionals.
3. Select deployment thresholds according to clinical priorities. In stroke screening, reducing false negatives may be more important than achieving maximum accuracy.
4. Monitor model performance after deployment through continuous evaluation of recall, precision and calibration.
5. Collect additional patient data from multiple hospitals to improve generalisability and reduce population bias.
6. Establish governance processes to ensure responsible use of predictions and protection of patient information.

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